

BHAVETHRA BAABU

Boston, MA | bbaabu@clarku.edu | linkedin/bhavethrab24 | github/BhavethraBaabu

EDUCATION

Master of Science in Computer Science (STEM) | Clark University, Worcester, MA Aug 2024 – May 2026
Bachelor of Technology in Information Technology | Anna University, Chennai, India Aug 2018 – Jul 2022

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Go

Frontend: React, Next.js, Tailwind CSS

Backend: FastAPI, Node.js, REST APIs, Microservices

Databases: PostgreSQL, MongoDB, MySQL, Redis, Vector Databases

AI/LLM: OpenAI API, LangChain, RAG, Generative AI, LLMs, Prompt Engineering, Vector Search

Cloud & Infrastructure: AWS, Docker, Kubernetes, Distributed Systems

DevOps & Tools: CI/CD, Git, GitHub Actions, Linux, Terraform

Testing & Monitoring: PyTest, Jest, Playwright, Prometheus, Grafana

EXPERIENCE

Software Engineer Intern | SciQuel Feb 2026 – Apr 2026
Boston, MA

- Built **full-stack features with Next.js** and React for a live science publishing platform spanning 15+ disciplines, onboarding 50+ open-source contributors through a standardized component architecture.
- Optimized **PostgreSQL** query performance and backend data infrastructure using **Prisma ORM**, reducing article and profile metadata fetch latency by **30%** and improving platform responsiveness under concurrent user load.
- Architected and delivered a responsive **atomic design system** in **TypeScript** and **Tailwind CSS**, cutting frontend development cycle time by **30%** and ensuring cross-platform UI consistency across all article and contributor interfaces
- Maintained **CI/CD pipelines** via **GitHub Actions**, reviewed and merged **20+ pull requests** across the open-source codebase, and enforced **TypeScript linting** and **testing standards** — achieving **production stability** across all deployments.

Software Engineer | LTIMindtree Jul 2022 – Jul 2024
Bengaluru, India

- Built and maintained **cloud-native** AWS deployment automation across 12 BNY Mellon MFT servers and DMZ middleware layers, enabling reliable software delivery and achieving **production-grade** system availability with zero unplanned downtime.
- Automated **MFT pipeline failure detection** using custom **scheduled jobs** and **alerting scripts**, eliminating manual monitoring and enabling real-time incident response.
- Built and deployed a **full observability stack** (Prometheus + Grafana) across production services, reducing mean time to resolution and empowering on-call engineers to debug incidents autonomously.
- Designed and maintained **RESTful APIs** for the Unitrax SaaS platform, implementing **JWT authentication**, role-based access control, and rate limiting to support secure, scalable backend integration for hundreds of enterprise clients

PROJECTS

CampusPulse Clark — Full-Stack AI Platform github.com/BhavethraBaabu/campuspulse
Next.js · TypeScript · Python · Groq · Qdrant · Supabase

- Built a **production-grade** AI assistant using **RAG**, indexing 285 university web pages into Qdrant with sentence-transformers and serving natural language queries through Groq Llama 3.1, achieving **real-time** responses with **sub-2s latency**.
- Developed and deployed **10+ full-stack features** including **AI chat**, **voice input**, **course recommendations**, **GPA calculator**, and a real-time **analytics dashboard** using **Next.js 14**, **TypeScript**, **Tailwind CSS**, and **Supabase**.
- Implemented secure authentication with **Google OAuth**, row-level security, and PostgreSQL; deployed a **production-ready cloud application** on Vercel with automated CI/CD, supporting concurrent users and scalable AI-powered search.

COINPULSE — Real-Time Crypto Intelligence Platform github.com/BhavethraBaabu/coinpulse
Next.js · TypeScript · WebSocket · PostgreSQL · Vercel

- Built a real-time market data pipeline ingesting live price feeds from Binance and CoinGecko via WebSockets, processing position updates and **portfolio P&L across 100+ assets** — mirroring core trade capture and reconciliation workflows.
- Architected a **React Server Components + Suspense** streaming UI with **Next.js App Router**, reducing **time-to-interactive by 35%** compared to client-side polling approaches and improving perceived performance under live market data load
- Designed and integrated 5 backend-driven modules — **portfolio tracking**, **paper trading**, **alerts**, and **market analytics** — mirroring trade capture and position management workflows across a unified financial data platform.